

Daily Report – Phase 2: Advanced Conditional Access & MFA (Zero Trust)

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Company: Consultim-IT

Objective

The objective of this session was to implement and validate **advanced Conditional Access policies** in Microsoft Entra ID to enforce **Multi-Factor Authentication (MFA)** and simulate a **Zero Trust security model**.

Note:

All configurations and administrative actions in this phase were performed using the **MOD Administrator** account (shared tenant), not the **Malek user account**.

1. Conceptual Understanding (Theory + Practice)

Before implementation, I focused on understanding the principles of **Zero Trust security**, especially:

“Never trust, always verify”

I learned that:

- Authentication must be continuously validated
- Access decisions depend on multiple conditions (identity, location, risk)
- Conditional Access is the core enforcement mechanism in Microsoft Entra

This theoretical understanding helped guide the practical configuration.

2. Preparation – Safe Testing Approach

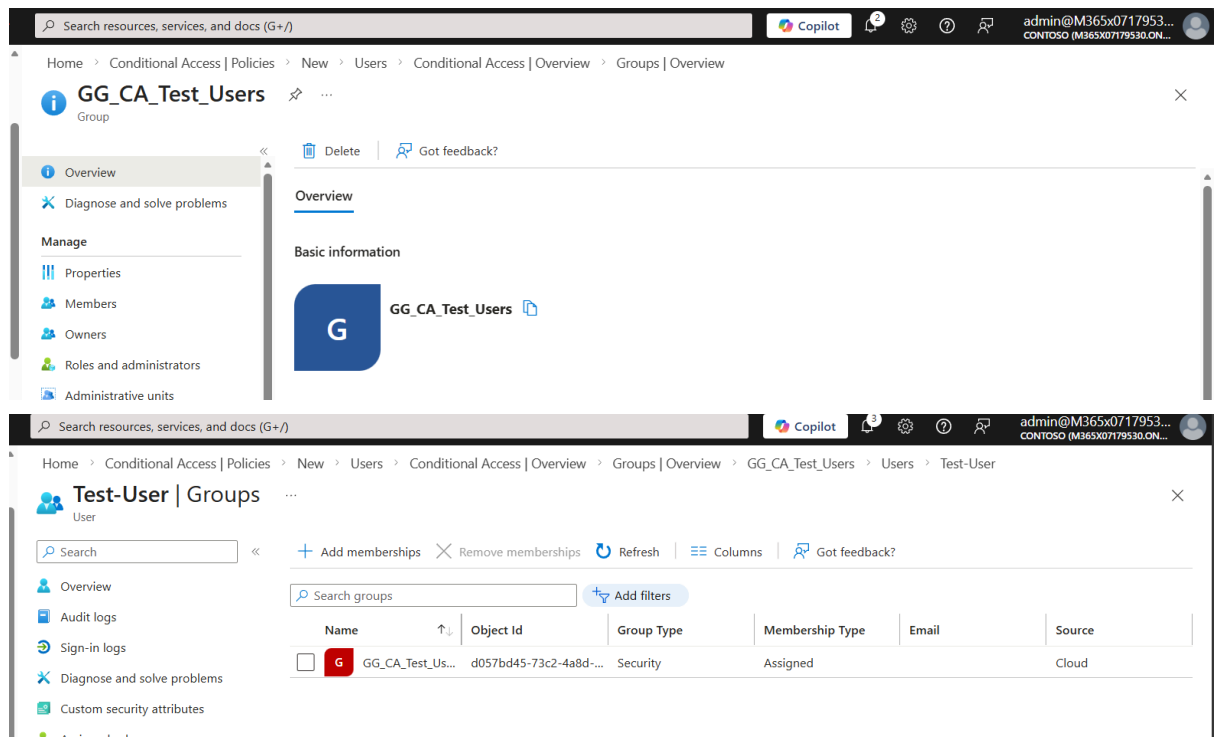
To avoid impacting the shared tenant, I applied best practices:

- Created a **dedicated test group**:
GG_CA_Test_Users
- Added a **single test user**
- Avoided applying policies to all users or administrators

Justification

This ensures:

- Safe testing
- No risk of locking accounts
- Scalable and professional configuration



3. Conditional Access Policies Implementation

I created four structured policies:

CA01 – Require MFA

- Users: Test group
- Cloud apps: All
- Control: Require MFA

Purpose

Ensure all access requires strong authentication.

Search resources, services, and docs (G+)

Home > Conditional Access | Policies > New > Users > Conditional Access | Overview > Groups | Overview > GG_CA_Test_Users > Users > Test-User | Groups > Conditional Access | Policies

New

Conditional Access policy

Name * CA01 - Require MFA ✓

What does this policy apply to? Users and groups

Include Exclude

☐ None

☐ All users

☒ Select users and groups

☐ Guest or external users

☐ Directory roles

☒ Users and groups

Select

1 group

GG CA_Test_Users

Assignments

Users or agents (Preview)

Specific users included

Target resources

No target resources selected

Network NEW

Not configured

Conditions

0 conditions selected

Access controls

Grant

CA02 – Block Legacy Authentication

- Users: Test group
- Condition: Legacy authentication clients
- Control: Block access

Purpose

Prevent bypass of MFA through outdated protocols.

Search resources, services, and docs (G+)

Home > Conditional Access | Policies > New > Users > Conditional Access | Overview > Groups | Overview > GG_CA_Test_Users > Users

New

Conditional Access policy

Name * CA02 - Block Legacy Authentication ✓

User risk level is the likelihood that the user account is compromised.

Not configured

Assignments

Users or agents (Preview)

Specific users included

Target resources

No target resources selected

Network NEW

Not configured

Conditions

0 conditions selected

Access controls

Block access

Sign-in risk

Sign-in risk level is the likelihood that the sign-in session is compromised.

Not configured

Insider risk

Insider risk assesses the user's risky data-related activity in Microsoft Purview Insider Risk Management.

Not configured

Device platforms

Not configured

Locations

Not configured

Client apps

Not configured

Client apps

Control user access to target specific client applications not using modern authentication. [Learn more](#)

Configure

Yes No

Select the client apps this policy will apply to

Modern authentication clients

☒ Browser

☒ Mobile apps and desktop clients

Legacy authentication clients

☒ Exchange ActiveSync clients

☒ Other clients

CA03 – MFA Outside Trusted Location

- Users: Test group
- Condition: Any location except trusted “**tunisia**”
- Control: block access

Purpose

Add adaptive security based on location.

New

Conditional Access policy

Name *
CA03 - MFA Outside Trusted Location

Assignments

Users or agents (Preview)
Specific users included

Target resources
No target resources selected

Network
Any network or location

Conditions
1 condition selected

Access controls

Grant
0 controls selected

User risk
User risk level is the likelihood that the user account is compromised.
Not configured

Sign-in risk
Sign-in risk level is the likelihood that the sign-in session is compromised.
Not configured

Insider risk
Insider risk assesses the user's risky data-related activity in Microsoft Purview Insider Risk Management.
Not configured

Device platforms
Not configured

Locations
Any network or location

Client apps

Configure
Yes No

- Include Exclude
- Any network or location
 - All trusted networks and locations
 - All Compliant Network locations
 - Selected networks and locations

To create a Conditional Access policy ensuring your tenant's members are coming from their compliant network, make sure Global Secure Access (GSA) is deployed and Adaptive Access Signaling in GSA is enabled in your tenant. Learn more on how to enable GSA Adaptive Access Signaling.

'Locations' condition is moving! Locations will become the 'Network' assignment with a new Global Secure Access capability of 'All Compliant'.

Search resources, services, and docs (G+)

Copilot

admin@M365x0717953...
CONTOSO (M365X07179530.ON...

Home > Conditional Access | Policies > New > Users > Conditional Access

Conditional Access | Named locations

Microsoft Entra ID

Overview

Policies

Deleted Policies

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

+ Countries location

+ IP ranges location

Configure multifactor authentication trusted IPs

Refresh

Preview features

...

Named locations are used by Microsoft Entra security reports to reduce false positives and by Microsoft Entra Conditional Access policies. Named Locations that are marked Trusted or configured in Conditional Access Policies cannot be deleted. Learn more

All Named Locations

Deleted Named Locations

Search by name

Location type: All types

Trusted type: All types

Reset filters

6 named locations found

Name	Location type	Trusted	Conditional Access...	Creation date	Modified date	
ME Services	IP ranges	Yes	Not configured...	3/16/2026, 6:08...	3/16/2026, 6:08...	...
Provisioning Workers	IP ranges	Yes	Not configured...	3/16/2026, 6:08...	3/16/2026, 6:08...	...
Stockpiling Workers	IP ranges	Yes	Not configured...	3/16/2026, 6:08...	3/16/2026, 6:08...	...
Teams Workers	IP ranges	Yes	Not configured...	3/16/2026, 6:08...	3/16/2026, 6:08...	...
Tunisia	Countries (IP)		Country factor	3/27/2026, 4:51...	3/27/2026, 4:51...	...
VPN and SAW IP Ranges	IP ranges	No	Not configured...	3/17/2026, 2:53...	3/17/2026, 2:53...	...

[Learn more](#)

Name *

CA03 - MFA Outside Trusted Location

Assignments

Users or agents (Preview)

[Specific users included](#)

Target resources

[No target resources selected](#)

Network **NEW**

[Any network or location and 1 excluded](#)

Conditions

Configure

Yes

No

Include

Exclude

Select the locations to exempt from the policy



All trusted networks and locations



All Compliant Network locations



Selected networks and locations

Select

Tunisia

Tunisia



CA04 – Block High Risk Sign-in

- Users: Test group
- Condition: High sign-in risk
- Control: Block access

Purpose

Automatically block suspicious login attempts.

The screenshot shows the 'New' Conditional Access policy configuration page in the Microsoft Entra ID portal. The policy is named 'CA04 - Block High Risk Sign-in'. It is configured with the following settings:

- Users or agents:** Specific users included
- Target resources:** No target resources selected
- Sign-in risk:** Not configured
- Insider risk:** Not configured

The 'Sign-in risk' configuration panel is open, showing the following options:

- Configure:** Yes (selected), No
- Sign-in risk level:** High (selected), Medium, Low, No risk

4. Testing & Validation (Step-by-Step)

I performed multiple test scenarios to validate each policy.

Test 1: Normal Login with MFA

- Logged in using test user (Incognito mode)
- MFA prompt appeared
- Access granted after validation



Se connecter

Continuer vers Microsoft Entra

test.user@M365x07179530.onmicrosoft.com

Vous n'avez pas encore de compte ? [Créez-en un !](#)

[Votre compte n'est pas accessible ?](#)

Suivant



← test.user@M365x07179530.onmicrosoft.com

Nous allons essayer



50

Entrez le numéro indiqué dans l'application pour approuver la demande de connexion.



test.user@M365x07179530.onmicrosoft.com

✓ Authenticator Added



Vous pouvez maintenant utiliser Microsoft Authenticator pour approuver les connexions, obtenir des codes uniques, etc.

Il s'agit maintenant de votre méthode de connexion par défaut.



Test 2: Login Without Completing MFA

- Attempted login
- Did not approve MFA

Result: Access blocked

Test 3: Access from Different Network

- Changed network, ip, location by **VPN "Amsterdam"**
- Block access



Connected



Amsterdam
Netherlands





Se connecter

Continuer vers Microsoft Entra

test.user@M365x07179530.onmicrosoft.com

Vous n'avez pas encore de compte ? [Créez-en un !](#)

[Votre compte n'est pas accessible ?](#)

Suivant

https://login.microsoftonline.com/common/login

ovo Gmail YouTube Google >رائط POO-TD-.pdf Under.the.Skin.2013 AD intune



Se connecter

Désolé, nous rencontrons des problèmes pour vous connecter.

AADSTS900561: The endpoint only accepts POST requests. Received a GET request.

Détails du dépannage

Si vous contactez votre administrateur, envoyez-lui ces informations.

[Copier les informations dans le presse-papiers](#)

Request Id: 479e25f2-9e22-4275-b282-60df82146d00

Correlation Id: cbded0a-5cef-4832-901c-5b94d155487d

Timestamp: 2026-03-27T17:25:54Z

Message: AADSTS900561: The endpoint only accepts POST requests. Received a GET request.

Signaler les erreurs de connexion pour la révision : [Activer les indicateurs](#)

Si vous prévoyez d'obtenir de l'aide pour ce problème, activez les indicateurs et essayez de reproduire l'erreur dans un délai de 20 minutes. Les événements avec indicateur rendent les diagnostics disponibles et sont portés à l'attention de l'administrateur.

Test 4: Verification in Sign-in Logs

- Navigated to:
Entra → Monitoring → Sign-in logs
- Verified:
 - Login status (success / failure)
 - MFA requirement
 - Applied Conditional Access policies

Sign-in events

Download

Export data Settings

Troubleshoot

Refresh

Manage view

Got feedback?

Want to switch back to the legacy sign-in logs experience? Click here to leave the preview.

Add filter

Show dates as: UTC

Date range: Last 24 hours

Reset filters

User sign-ins (interactive)

User sign-ins (non-interactive)

Service principal sign-ins

Managed identity sign-ins

Date ↓	Request ID	User principal name	Application	Status	IP address	Resource
2026-03-27T17:21:14Z	2afa9376-0cc9-43e8-9990-674e1...	admin@m365x0717953...	Azure Portal	Success	2a10:1fc0:c0:216:3cffe...	Azure Resource
2026-03-27T17:16:53Z	a5ea99bd-83ed-42a4-a025-c6d7...	admin@m365x0717953...	Azure Portal	Success	209.127.202.113	Azure Resource
2026-03-27T17:13:58Z	9da5d3a4-a53a-4480-9ff1-8d66f...	test.user@m365x07179...	Azure Portal	Success	197.240.49.249	Azure Resource
2026-03-27T17:13:54Z	e918ff7d-41af-4a5f-bab7-c69aa8...	test.user@m365x07179...	Azure Portal	Interrupted	197.240.49.249	Azure Resource
2026-03-27T16:13:38Z	47d4d8cd-f5a5-4c95-be7a-9caf5...	admin@m365x0717953...	Azure Portal	Success	197.240.49.249	Azure Resource
2026-03-27T16:10:40Z	e3015c74-6c3f-41f2-b493-08fa8f...	amine.benyoussef@m3...	Azure Portal	Success	68.183.109.33	Azure Resource
2026-03-27T16:10:35Z	f3b08662-56ef-4416-9c48-3617d...	amine.benyoussef@m3...	Azure Portal	Interrupted	68.183.109.33	Azure Resource
2026-03-27T16:08:15Z	b9e001b5-b895-4461-8ed5-c35e...	amine.benyoussef@m3...	Azure Portal	Failure	68.183.109.33	Azure Resource

Sign-in events

Download

Export data Settings

Troubleshoot

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User sign-ins (interactive)

User sign-ins (non-interactive)

Service principal sign-ins

Managed identity sign-ins

Status	IP address	Resource	Resource ID	Conditional ac...	User	Location
Success	2a10:1fc0:c0:216:3cffe...	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	MOD Administrator	, , NL
Success	209.127.202.113	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	MOD Administrator	Amsterdam, Noord-Holl...
Success	197.240.49.249	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	Test-User	Tunis, Tunis, TN
Interrupted	197.240.49.249	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	Test-User	Tunis, Tunis, TN
Success	197.240.49.249	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	MOD Administrator	Tunis, Tunis, TN
Success	68.183.109.33	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	Amine Ben Youssef	Broomfield, Colorado, US
Interrupted	68.183.109.33	Azure Resource Manager	797f4846-ba00-4fd7-ba...	Success	Amine Ben Youssef	Broomfield, Colorado, US

5. Key Observations

- MFA is successfully enforced through Conditional Access
- Access is blocked when verification fails
- Policies are dynamically applied based on conditions
- Sign-in logs provide full visibility of authentication events

6. Best Practices Applied

- Used **group-based assignment** instead of individual users
 - Tested policies before applying globally
 - Avoided impacting admin accounts
 - Used Incognito mode for clean testing sessions
-

Results

- ✓ Advanced Conditional Access policies successfully implemented
 - ✓ MFA enforced across scenarios
 - ✓ Legacy authentication blocked (log-based verification)
 - ✓ Location-based access control tested
 - ✓ Sign-in logs used for monitoring and validation
 - ✓ Zero Trust model effectively simulated
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